

Ex. 55. Mr. Bhaskar is on tour and he has ₹ 360 for his expenses. If he exceeds his tour by 4 days, he must cut down his daily expenses by ₹ 3. For how many days is Mr. Bhaskar on tour?

Sol. Suppose Mr. Bhaskar is on tour for x days. Then,

$$\frac{360}{x} - \frac{360}{x+4} = 3 \Leftrightarrow \frac{1}{x} - \frac{1}{x+4} = \frac{1}{120} \Leftrightarrow x(x+4) = 4 \times 120 = 480$$

$$\Leftrightarrow x^2 + 4x - 480 = 0 \Leftrightarrow (x+24)(x-20) = 0 \Leftrightarrow x = 20.$$

Hence, Mr. Bhaskar is on tour for 20 days.

Ex. 56. A railway half-ticket costs half the full fare. But the reservation charge on half-ticket is the same as that on full ticket. One reserved first-class ticket for a journey between two stations is ₹ 525 and the cost of one full and one half reserved first-class tickets is ₹ 850. What is the reservation charge? (Bank P.O., 2008)

Sol. Let the full fare be ₹ x and the reservation charge per ticket be ₹ y .

$$\text{Then, } x + y = 525 \quad \dots(i)$$

$$\text{And, } \frac{3x}{2} + 2y = 850 \text{ or } 3x + 4y = 1700 \quad \dots(ii)$$

Multiplying (i) by 3 and subtracting from (ii), we get: $y = 125$.

Hence, reservation charge = ₹ 125.

Ex. 57. Tom reads at an average rate of 30 pages per hour, while Jane reads at an average rate of 40 pages per hour. If Tom starts reading a novel at 4: 30 and Jane begins reading an identical copy of the same book at 5: 20, then at what time will they be reading the same page? (M.A.T., 2007)

Sol. Suppose they are reading the same page x hours after 5: 20.

$$\text{Time from 4: 30 to 5: 20} = 50 \text{ min} = \frac{5}{6} \text{ hr.}$$

$$\text{Then, } 30\left(x + \frac{5}{6}\right) = 40x \Leftrightarrow 30x + 25 = 40x \Leftrightarrow 10x = 25 \Leftrightarrow x = 2.5.$$

Hence, Tom and Jane will be reading the same page $2\frac{1}{2}$ hrs i.e., 2 hr 30 min after 5: 20 i.e., at 7: 50.

Ex. 58. In an objective examination of 90 questions, 5 marks are allotted for every correct answer and 2 marks are deducted for every wrong answer. After attempting all the 90 questions a student got a total of 387 marks. Find the number of questions that he attempted wrong. (M.A.T. 2007)

Sol. Let the number of questions attempted correctly be x .

$$\text{Then, number of questions attempted wrong} = (90 - x)$$

$$\therefore 5x - 2(90 - x) = 387 \Leftrightarrow 7x = 387 + 180 = 567 \Leftrightarrow x = 81.$$

$$\text{Hence, number of questions attempted wrong} = (90 - 81) = 9.$$

Ex. 59. Kiran had 85 currency notes in all, some of which were of ₹ 100 denomination and the remaining of ₹ 50 denomination. The total amount of all these currency notes was ₹ 5000. How much amount did she have in the denomination of ₹ 50?

Sol. Let the number of 50-rupee notes be x .

$$\text{Then, the number of 100-rupee notes} = (85 - x).$$

$$\therefore 50x + 100(85 - x) = 5000 \Leftrightarrow x + 2(85 - x) = 100 \Leftrightarrow x = 70.$$

$$\text{So, required amount} = ₹ (50 \times 70) = ₹ 3500.$$

Ex. 60. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. Find the number of people who like both coffee and tea. (P.C.S., 2007)

Sol. Let A and B represent the set of people who like coffee and tea respectively.

$$\text{Then, } n(A) = 37, n(B) = 52, n(A \cup B) = 70.$$

$$\begin{aligned} \therefore n(A \cap B) &= n(A) + n(B) - n(A \cup B) \\ &= 37 + 52 - 70 = 89 - 70 = 19. \end{aligned}$$

Hence, number of people who like both coffee and tea = 19.

